PERT Chart

Remediation plan in Security operations Centre or Structure incidence response to manage project risks

Paul Mutunda – Course Lecture © 2024 for KMU

Task 1 – Identify Remediation activities

1. Patching vulnerabilities in acted systems
2. Securing compromised user accounts
3. Rolling back or restoring impacted applications or data
4. Implementing additional security controls
5. Revising policies and procedures to prevent recurrences
6. Conducting post-remediation audits and testing

Task 1 – Determine dependences

- Identify dependences and map them in the diagram

Task 1 – Identify Remediation activities

1. For each remediation activity, estimate the following;

- Optimistic Time (O)
- Pessimistic Time (P)
- Most likely Time (M)

Task 1 – Create the PERT Chart

- Draw nodes representing milestones or completion points
- Connect with arrows representing tasks following their dependencies
- Assign expected times using the formula **te = to + 4tm + tp/6** to each task

Task 1 – Identify the Critical path and the variance and SD

- This path represents the longest sequence of dependent tasks that determine the shortest possible remediation time. Prioritize resources here to avoid delays
- Calculate the variance and the standard deviation to assess uncertainty and risk for each activity and the overall remediation duration

Task 1 – Monitor and adjust

- As remediation progresses, update the PERT chart with crucial durations and re-assess timelines and risks

Task 1 – Identify Remediation activities

Activity	O	P	M	Dependences
Identify compromised systems	1	3	2	None
Patch vulnerabilities	2	6	4	Identify compromised vulnerabilities
Secure User Accounts	1	4	2	Patch vulnerabilities
Restore data	3	8	5	Patch vulnerabilities
Implement additional controls	2	5	3	Secure User Accounts
Revise Policies	1	3	2	Identify compromised systems
Final audit and testing	2	4	3	Implement additional controls, restore data

Task 2 – system hardening as a remediation plan

Task	Optimistic	Most likely	Pessimistic
Root cause analysis	1 day	2 days	4 days
Patch development	2 days	4 days	6 days
Pact testing	1 day	3 days	5 days
Schedule Maintenance	1 day	2 days	3 days
Deploy patch	2 days	2 days	4 days
Validate remediation	1 day	2 days	3 days
Update documentation	1 day	1 day	2 days
Notify stakeholders	0.5 days	1 day	15 days

Task 1 – Monitor and adjust

- Visualize this information using a PERT CHART to guide your team in the department on the remediation plan outlined
- $$\text{Time} = \frac{\text{Optimistic} + 4 * \text{Most Likely} + \text{Pessimistic}}{6}$$
- Find the critical path which dictates the shortest time to complete this remediation project plan.
- Report on how you will amange resources to avoid delays on the critical path
- Adjust timelines and resource allocation needed.